

Level-Up Class: Strings & Pattern Printing

Introduction

Students, welcome to the next level

Today we extend what you learned about loops.

- Talk to letters using loops
- Play with strings
- Draw patterns using code

Relax, read slowly, imagine the output, then code.

Part 1: String Looping — Talking to Letters

A **string** is a collection of characters. Python can give us one character at a time using a loop.

Question 1: Say Hello to Each Letter

Task: Ask the user for a word and print each letter with a greeting.

Example Output:

```
Hello P
Hello y
Hello t
Hello h
Hello o
Hello n
```

Solution:

```
1 word = input("Enter a word: ")
2
3 for ch in word:
4     print("Hello", ch)
```

Question 2: Count How Many Times 'a' Appears

Task: Count how many times the letter a appears in a word.

Solution:

```
1 word = input("Enter a word: ")
2
3 count = 0
4
5 for ch in word:
6     if ch == "a":
7         count = count + 1
8
9 print("Total 'a':", count)
```

Question 3: Remove Spaces from a Sentence

Task: Print the sentence again but without spaces.

Solution:

```
1 sentence = input("Enter a sentence: ")
2
3 for ch in sentence:
4     if ch != " ":
5         print(ch, end="")
```

Question 4: Reverse a Word (Using Loop)

Task: Reverse a word manually using a loop.

Solution:

```
1 word = input("Enter a word: ")
2
3 reverse_word = ""
4
5 for ch in word:
6     reverse_word = ch + reverse_word
7
8 print("Reversed:", reverse_word)
```

Part 2: Pattern Printing — Drawing with Code

Patterns are drawings made using loops. Stars (*) are our pencil

Pattern 1: Growing Stars

Output:

```
*  
**  
***  
****  
*****
```

Code:

```
1 for i in range(1, 6):  
2     print("*" * i)
```

Pattern 2: Shrinking Stars

Output:

```
*****  
****  
***  
**  
*
```

Code:

```
1 for i in range(5, 0, -1):  
2     print("*" * i)
```

Pattern 3: Number Ladder

Output:

```
1  
22  
333  
4444  
55555
```

Code:

```
1 for i in range(1, 6):  
2     print(str(i) * i)
```

Pattern 4: Counting Triangle

Output:

```
1  
12  
123  
1234  
12345
```

Code:

```
1 for i in range(1, 6):  
2     for j in range(1, i + 1):  
3         print(j, end="")  
4     print()
```

Pattern 5: Center Star (Mini Diamond)

Output:

```
  *  
 ***  
*****
```

Code:

```
1 for i in range(1, 4):  
2     spaces = 3 - i  
3     stars = 2 * i - 1  
4     print(" " * spaces + "*" * stars)
```

Final Thought

Programming is not just typing code.

Programming means:

- Thinking

- Repeating
- Playing with logic

A loop simply means: **Do this again until I say stop.**

Mini Challenge (Practice at Home)

1. Print your name letter by letter
2. Count vowels in your name
3. Print the following pattern:

```
@
@@
@@@
@@@@
```

4. Reverse your name using a loop